ACCELERATION

LEARNING GOALS

Students will:

- Define acceleration.
- Calculate average acceleration from vector velocities.

DEFINITION

Acceleration is the rate of change of velocity.

1. Write the equation that agrees with this definition:

$$\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t}$$

velocity changes when speed or direction changes

AVERAGE ACCELERATION

CONSIDER THE FOLLOWING:

A bird is flying at 3.4 m/s [S]. Over the next 1.5 s, it turns until it is flying at 3.4 m/s [W].

2. As a group, discuss: What is the direction of its average acceleration?
   a) west
   b) south
   c) south-west
   d) north-west
   e) nonsense, there is no acceleration because the speed is constant

3. Fill in the following quantities:
   $\vec{v_1} =$ 3.4 m/s [S] $\vec{v_2} =$ 3.4 m/s [W] $\Delta t =$ 1.5 s
4. You know that

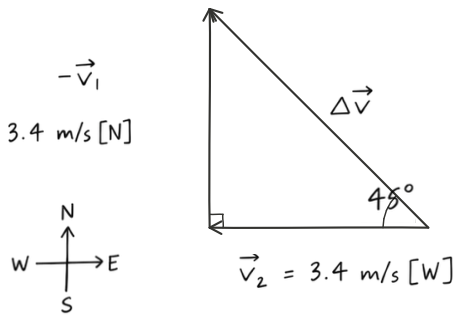
$$\vec{a_{ave}} = \frac{\Delta\vec{v}}{\Delta\vec{t}} = \frac{\vec{v_2} - \vec{v_1}}{\Delta t}$$

$$\vec{a_{av}} = \frac{3.4\frac{m}{s}[W] - 3.4\ \frac{m}{s}\ [S]}{1.5s}$$

The numerator is vector subtraction – change it to vector addition (add the negative):

$$\vec{a}_{av} = \frac{3.4 \text{ m/s [W]} + 3.4 \text{ m/s [N]}}{1.5 \text{ s}}$$

5. Now draw a vector diagram to do the vector addition in the numerator. Use trig (sine law or cosine law) to find the magnitude and direction of the resultant ($\Delta\vec{v}$).



$$|\Delta\vec{v}| = \sqrt{3.4^2 + 3.4^2}$$

$$= 4.81 \text{ m/s}$$

$$\theta = \tan^{-1}(3.4/3.4) = 45°$$

Δv = 4.8 m/s [NW]

6. Now complete the solution by dividing by the time.
7. Consider the direction of your answer. Explain the direction in terms of the physics involved.

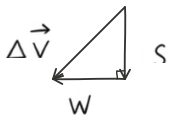
6) $\vec{a}_{av} = \frac{4.81\ m/s\ [NW]}{1.5\ s} = 3.2\ m/s^2\ [NW]$

north reduces the southward velocity, and west builds the westward velocity.

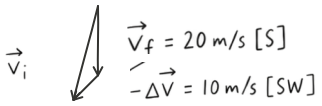
8. Say a car is travelling east at 60 km/h and executes a turn after which it is travelling 60 km/h [S]. In what direction is the acceleration?

$\Delta\vec{V} = 60\ km/h\ [S] + 60\ km/h\ [W]$

$\therefore \vec{a}_{av}$ is [SW]



9. You know that $\Delta\vec{v} = \vec{v}_f - \vec{v}_i$.
   a. Rewrite the equation to solve for initial velocity. $\vec{V}_i = \vec{V}_f - \Delta\vec{V}$
   b. If final velocity is 20 m/s [S] and the change in velocity is 10 m/s [NE], sketch a vector diagram (based on a.) that would allow you to find initial velocity.



$\vec{V}_i = \vec{V}_f + (-\Delta\vec{V})$

$\vec{V}_i = 28\ m/s\ [S15°W]$

5. 4.8 m/s [NW] 6. 3.2 m/s^2 [NW] 8. SW

working on following pages

PRACTICE

For each of the following...

- Draw the original vectors
- Write out the equations and vector statements involved
- Draw the correct vector diagram to represent the equation
- Use the vector diagram to find the answer

1. A bird travelling 3.0 m/s NW turns and heads south at the same speed. Determine its change in velocity. (5.5 m/s [S 22.5° E])

2. After completing a turn, a cyclist is travelling east at 25 km/h. If her change in velocity was 15 km/h south, determine her initial velocity. (29 km/h [E 31° N])

3. A go cart rounding a curve goes from an initial velocity of 60 km/h [S] to 40 km/h [E]. If the curve takes 2.0 s, determine the cart's acceleration. (36km/h/s [E56°N])

4. A car, moving initially at 32 km/h [N], turns a corner and continues at 32 km/h [W]. The turn takes 3.0 s to complete. Find
   a. the change in velocity (45 km/h [SW])
   b. the average acceleration during the turn. (15 km/h [SW])

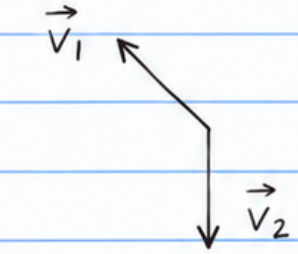
5. A fighter jet comes out of a curve traveling 750 km/h [N]. During the curve, it experienced an acceleration of 3 g's (three times acceleration due to gravity) [SW] for a duration of 2.5 s. What was the jet's initial velocity? (265 m/s [N 11° E]

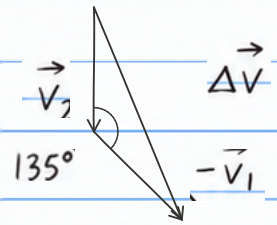
1.6 2D acceleration practice

1) bird: NW to S

$\vec{v}_1 = 3.0$ m/s [NW] $\quad \vec{v}_2 = 3.0$ m/s [S]

original





$$\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$$

$$|\Delta\vec{v}| = \sqrt{3^2 + 3^2 - 2(3)(3)\cos 135°}$$
$$= 5.543 \text{ m/s}$$

$$\frac{\sin\theta}{3.0} = \frac{\sin 135°}{5.543}$$

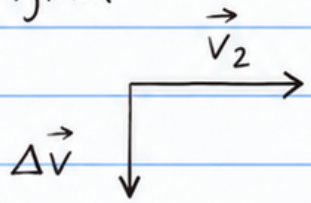
$$\theta = \sin^{-1}(0.3827)$$
$$= 22.5° \text{ east of south}$$

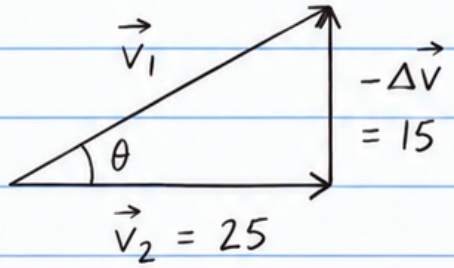
$\Delta\vec{v} = 5.5$ m/s [S 22.5° E]

2) cyclist: find initial velocity

$\vec{v}_2 = 25$ km/h [E] $\quad \Delta\vec{v} = 15$ km/h [S]

original





$$\vec{v}_1 = \vec{v}_2 - \Delta\vec{v}$$
$$= \vec{v}_2 + (-\Delta\vec{v})$$

$$|\vec{v}_1| = \sqrt{25^2 + 15^2}$$
$$= 29.15 \text{ km/h}$$

$$\theta = \tan^{-1}(15/25)$$
$$= 31° \text{ north of east}$$

$\vec{v}_1 = 29$ km/h [E 31° N]

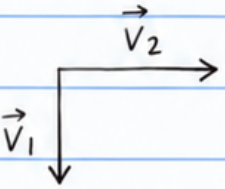
southward change cancels the northward part

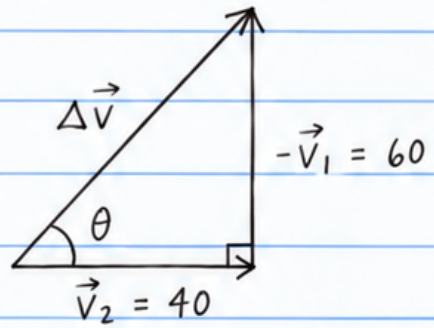
1.6 acceleration practice

3) go cart

$\vec{v}_1 = 60$ km/h [S] $\quad \vec{v}_2 = 40$ km/h [E] $\quad \Delta t = 2.0$ s

original





$\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$

$|\Delta\vec{v}| = \sqrt{40^2 + 60^2}$

$= 72.11$ km/h

$\theta = \tan^{-1}(60/40) = 56°$

$\vec{a}_{av} = \frac{72.11}{2.0}$ [E56°N]

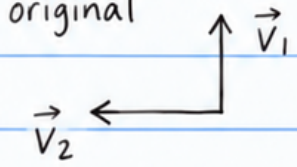
$\vec{a}_{av} = 36$ (km/h)/s [E56°N]

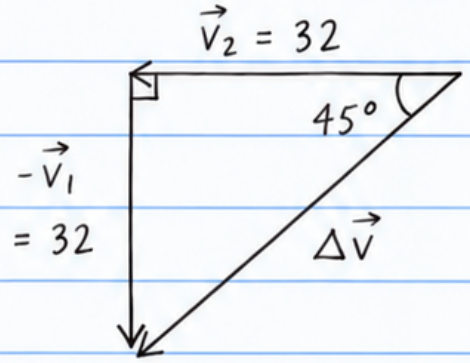
$= 10$ m/s^2 [E56°N]

4) car: N to W

$\vec{v}_1 = 32$ km/h [N] $\quad \vec{v}_2 = 32$ km/h [W] $\quad \Delta t = 3.0$ s

original





a) $\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$

$|\Delta\vec{v}| = \sqrt{32^2 + 32^2}$

$= 45.25$ km/h

$\theta = \tan^{-1}(32/32) = 45°$

$\Delta\vec{v} = 45$ km/h [SW]

b) $\vec{a}_{av} = \frac{45.25 \text{ km/h}}{3.0 \text{ s}}$

$\vec{a}_{av} = 15$ (km/h)/s [SW]

$= 4.2$ m/s^2 [SW]

1.6 acceleration practice

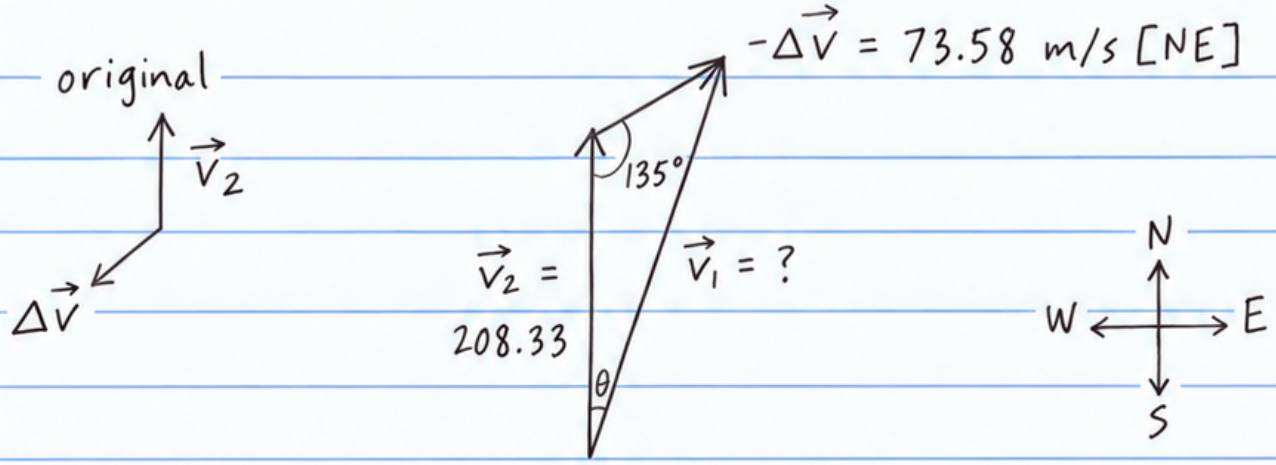
5) fighter jet: find initial velocity

$\vec{v}_2 = 750 \text{ km/h [N]} = \frac{750}{3.6} = 208.33 \text{ m/s [N]}$

$\vec{a}_{av} = 3g \text{ [SW]} = 29.43 \text{ m/s}^2 \text{ [SW]} \quad \Delta t = 2.5 \text{ s}$

$\Delta\vec{v} = \vec{a}_{av}\,\Delta t = (29.43)(2.5) = 73.58 \text{ m/s [SW]}$

$\vec{v}_1 = \vec{v}_2 - \Delta\vec{v} = \vec{v}_2 + (-\Delta\vec{v})$



use cosine law

$|\vec{v}_1|^2 = 208.33^2 + 73.58^2 - 2(208.33)(73.58)\cos 135°$

$= 70493$

$|\vec{v}_1| = \sqrt{70493} = 265.5 \text{ m/s}$

$\frac{\sin\theta}{73.58} = \frac{\sin 135°}{265.5}$

$\theta = \sin^{-1}(0.1959)$

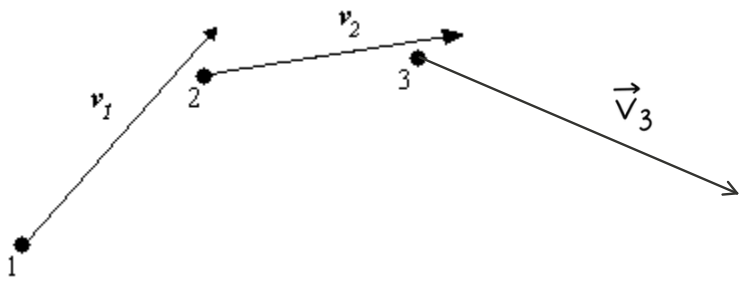
$= 11.3°$ east of north

$\therefore \vec{v}_1 = 2.7 \times 10^2 \text{ m/s [N11°E]}$

the jet was faster before the curve and had an eastward component before the SW acceleration

6. The diagram below a racing car's position at three equally spaced points in time. It also shows the car's instantaneous velocity vectors at points 1 and 2. If the acceleration of the car is uniform, determine, using an accurate vector scale diagram, the instantaneous velocity of the car at point 3.

   Scale: 1 cm = 5.0 m/s.



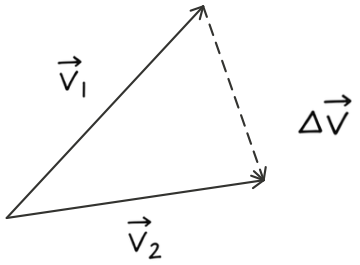
7. Read section 1.4 and do p 34 #1,2, p35#9

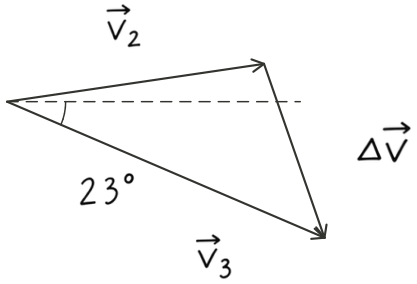
6) racing car

uniform acceleration: equal times give equal velocity changes

$\Delta\vec{V} = \vec{V}_2 - \vec{V}_1$ $\qquad$ $\vec{V}_3 = \vec{V}_2 + \Delta\vec{V}$

1) tails together $\qquad$ 2) add the same $\Delta\vec{V}$





scale: 1 cm = 5.0 m/s

$\Delta v_x = 17 - 13 = 4$ m/s

$\Delta v_y = 2.5 - 14 = -11.5$ m/s

$v_{3x} = 17 + 4 = 21$ m/s

$v_{3y} = 2.5 - 11.5 = -9$ m/s

measured length $\approx$ 4.6 cm

$|\vec{V}_3| = \sqrt{21^2 + 9^2}$

$= 22.8$ m/s

$\theta = \tan^{-1}(9/21)$

$= 23°$ below right

$\therefore \vec{V}_3 \approx 23$ m/s, 23° below right